

ML Interconnection Queue Intelligence for the U.S. Electric Grid

CEE 498 MLC — Spring 2026 Project Proposal

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1. Problem Statement and Motivation

The U.S. electrical grid faces a significant bottleneck in its transition to clean energy. As of the end of 2024, over 2,300 GW of generation capacity was actively waiting in interconnection queues for approval to connect to the grid, and historically more than 70% of those projects are eventually withdrawn. The average project now waits in excess of five years to complete the required interconnection impact studies. A critical challenge is that project developers have no reliable way to estimate, prior to filing, what network upgrade costs will be incurred or how long the study process will take. As a result, developers routinely commit millions of dollars in pre-development costs before receiving study results that render a project economically unviable.

This project proposes a machine learning platform trained on publicly available historical interconnection queue data to predict three project-level outcomes before a developer files an interconnection request: (1) the probability that a project will be withdrawn from the queue, (2) the likely category of required network upgrade costs (bucketed as \$0–1M, \$1–5M, \$5–20M, and \$20M+), and (3) the expected time in months to complete the study process. Reliable estimates of these outcomes at the site-evaluation stage would meaningfully improve developer decision-making and reduce wasteful investment in non-viable projects.

2. Project Format and Research Paper

This project satisfies both acceptable formats described in the course guidelines. It follows the *Implementation and Demo* format by reproducing the training methodology of a published research paper directly related to topics covered in CEE 498, applied to a civil engineering dataset. It also constitutes *Applying Methods Covered in Class* in that it applies several core course methods — logistic regression, linear regression, Gaussian process regression, neural networks, autoencoders, and physics-informed machine learning — to an infrastructure problem not previously worked on by the author.

Research Paper Being Reproduced

The primary methodology reproduced is from Raissi, Perdikaris, and Karniadakis (2019), “[Physics-Informed Neural Networks: A Deep Learning Framework for Solving Forward and Inverse Problems Involving Nonlinear Partial Differential Equations](#)”, published in the *Journal of Computational Physics*. This paper corresponds directly to the physics-informed machine learning topic covered in CEE 498. The central contribution of Raissi et al. is a neural network training framework in which a known governing law is embedded into the loss function as a physics residual term, rather than being learned purely from labeled data. The composite loss function takes the form:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{data}}(\theta) + \lambda \mathcal{L}_{\text{physics}}(\theta) \quad (1)$$

where $\mathcal{L}_{\text{data}}$ is the standard supervised loss over labeled training examples, $\mathcal{L}_{\text{physics}}$ penalizes viola-

tions of a known governing law evaluated at collocation points sampled from the input space, and λ is a weighting hyperparameter. The core finding of the paper is that this formulation produces more accurate and physically consistent predictions than unconstrained networks, with the largest gains observed in data-sparse regions where the governing law provides critical inductive bias.

What is being reproduced. This project reproduces the PINN training framework: the composite loss structure, collocation-point sampling, and the ablation study design comparing constrained versus unconstrained networks on both accuracy and physical plausibility. In the original paper the physics residual is derived from a partial differential equation. In this application, the governing constraint comes from FERC Order 2023, which establishes that required network upgrade costs cannot decrease as a project’s requested capacity (MW) increases, all else held equal — a monotonicity relationship grounded in transmission power-flow physics. The network is penalized during training whenever its cost predictions violate this rule across collocation points sampled from the (MW, congestion) feature subspace, directly replicating the PINN training structure. An ablation study in the spirit of Raissi et al. will compare the constrained and unconstrained models on accuracy and rate of physically implausible outputs.

3. Methodology

The full modeling pipeline applies the following methods, all covered in CEE 498.

Interpretable Baselines. Logistic regression is used as the baseline for withdrawal prediction (binary classification) and linear regression for timeline estimation (regression). These establish performance floors and provide coefficient-level interpretability over the engineered feature set, which includes queue position, ISO region, fuel type, requested MW capacity, voltage level, substation congestion proxy, and filing year.

Neural Network with Physics-Informed Constraint. A multi-layer perceptron (MLP) with dropout regularization serves as the primary predictive model for all three tasks and is the network to which the Raissi et al. physics constraint is applied. Standard training practices are employed: mini-batch stochastic gradient descent, batch normalization, early stopping on a held-out validation set, and an 80/10/10 train/validation/test split stratified by ISO and filing year to prevent temporal data leakage.

Gaussian Process Regression. GPR is applied to the cost estimation task to provide calibrated predictive uncertainty. Because interconnection data is unevenly distributed across substations, GPR produces uncertainty intervals that appropriately widen in data-sparse regions, giving practitioners honest confidence estimates alongside predictions.

Autoencoder. An autoencoder is trained over normalized queue records to produce a low-dimensional latent representation of each project. These embeddings are used for similarity retrieval, surfacing historically comparable projects alongside each prediction as concrete contextual reference cases.

4. Dataset Description

The training data is assembled entirely from publicly available regulatory sources. No proprietary data is required.

Primary Dataset. The [Lawrence Berkeley National Laboratory “Queued Up” 2025 Edition](#) compiles and cleans project-level interconnection queue records from all seven U.S. ISO/RTOs and 49 non-ISO utilities, representing approximately 97% of installed U.S. generating capacity. The freely downloadable Excel file contains: project identifier, ISO/RTO region, fuel type, requested MW

capacity, queue status (withdrawn, active, or operational), filing date, interconnection agreement status, and study phase duration metrics. After filtering for feature completeness, the estimated usable training set is 10,000–13,000 historical records spanning the early 2000s through 2024. Outcome labels for withdrawal prediction and timeline regression are directly available in this file.

Supplementary ISO Queue Exports. To obtain substation-level identifiers, voltage level, county, and queue position fields, per-ISO exports are joined using project identifiers:

- **MISO:** [MISO Generator Interconnection Queue](#) — live Excel export with historical records to the mid-2000s; includes substation name, county, voltage level, and study phase.
- **PJM:** [PJM New Services Queue](#) — covers active and historical records from both the legacy serial and current cycle-based queues.
- **CAISO:** [CAISO Public Queue Report](#) — active, withdrawn, and completed projects with deliverability status.

The outcome class distribution for withdrawal prediction is approximately 70% withdrawn and 30% approved or active. Stratified sampling and class-weight balancing will be applied to address this imbalance. The key engineered feature is a substation congestion proxy: the number of projects queued at the same point of interconnection as the target project.

5. Expected Outcomes, Hypotheses, and Evaluation Metrics

Hypotheses

1. Queue position and substation-level congestion will be the dominant predictors of both withdrawal probability and network upgrade cost, consistent with the power-flow intuition that projects at congested substations face larger required upgrades and are more likely to find the economics unfavorable upon receiving study results.
2. The MLP will outperform the logistic and linear regression baselines on all three tasks by learning nonlinear interactions between features — particularly between ISO region, fuel type, and MW capacity — that linear models cannot represent without explicit feature engineering.
3. Incorporating the physics-informed monotonicity penalty will improve cost prediction accuracy in data-sparse substation regions and reduce the rate of physically implausible predictions, consistent with the findings of Raissi et al. (2019).
4. GPR will produce better-calibrated uncertainty estimates for cost prediction than point estimates from the MLP alone, particularly at substations with fewer than 20 historical records.

Evaluation Metrics and Benchmarks

All models are evaluated on a held-out test set comprising 10% of the data, stratified by ISO and filing year to prevent temporal data leakage. Target benchmarks are based on comparable tabular classification tasks in the energy systems literature.

Task		Metric(s)	Baseline	Target
Withdrawal prediction		AUC-ROC, F1 (positive class)	Majority-class classifier (AUC = 0.50)	AUC $\geq$ 0.78
Cost bucket classification		Weighted F1, Accuracy	Frequency-weighted random ($\approx$ 0.50)	Weighted F1 $\geq$ 0.70
Timeline regression		MAE (months), R^2	Mean-prediction baseline	$R^2 \geq$ 0.55
GPR uncertainty		PICP at 90% CI	Uncalibrated point estimate	PICP $\geq$ 0.85
PINN study	ablation	Implausibility rate, Δ Accuracy	Unconstrained MLP	Reduced violations, no accuracy loss

6. Statement of Individual Responsibilities and Contributions

This project will be completed independently by Eshana Jain. All components — including data collection and preprocessing, feature engineering, model implementation and training (logistic regression, linear regression, MLP with physics-informed constraint, GPR, and autoencoder), evaluation, and preparation of the final report and presentation — are the sole responsibility of the author.

7. Key References

- [1] Raissi, M., Perdikaris, P., & Karniadakis, G.E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378, 686–707. [doi:10.1016/j.jcp.2018.10.045](https://doi.org/10.1016/j.jcp.2018.10.045)
- [2] Rand, J., et al. (2025). *Queued Up: 2025 Edition*. Lawrence Berkeley National Laboratory. emp.lbl.gov/queues
- [3] MISO Generator Interconnection Queue (dataset). misoenergy.org
- [4] PJM New Services Queue (dataset). [PJM Interconnection Data Table](#)
- [5] CAISO Public Queue Report (dataset). caiso.com/documents/publicqueueereport.xlsx
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- [8] Murphy, K. (2022). *Probabilistic Machine Learning: An Introduction*. MIT Press. github.com/probml/pml-book